

Traffic Vehicle Classification

Model evaluation report - validation and test performance of all 13 trained models

field	value
generated	2026-10-08 19:56:10
config	C:\Users\viroo\PycharmProjects\TrafficVehicle\configs\default.json
run fingerprint	1ba0f75a8c63be033d34e0cc95f1a600081d2d887b9763e64850f8adddd401
classes	8: ambulance, autobus, kamyun, kamyunet, minibus, savari, taxi, vanet
dataset split	930 train / 232 validation / 64 test
models	13 experiments, all 200 epochs
production model	ResNet18 fine-tuning (resnet18_fine_tuning)
production checkpoint sha256	8e1147b5ee5108737ac3a1ad92704e7af3e08215b6d6f81b78f3e55545a7c6d3
selection protocol	best validation macro-F1; test split used for reporting only

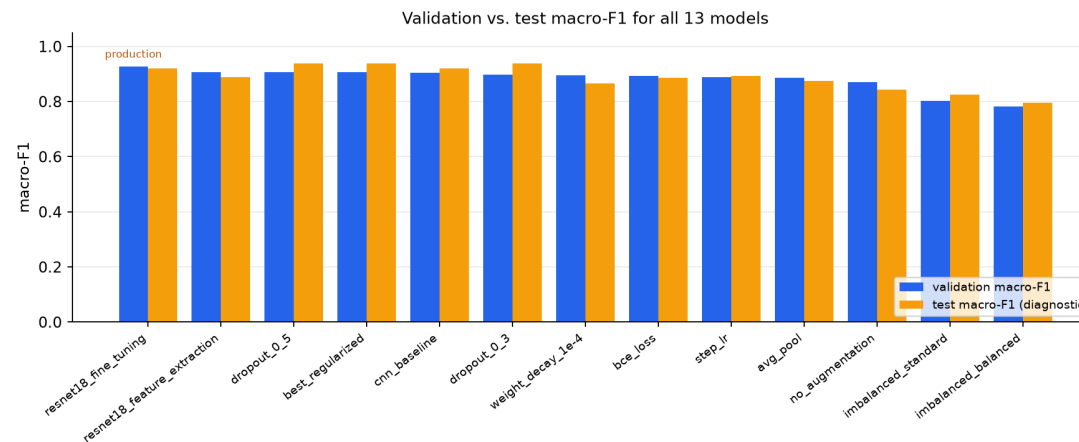
Key findings

- Highest validation macro-F1: **ResNet18 fine-tuning** (0.9274 at epoch 125).
- Selected production model: **ResNet18 fine-tuning** (validation macro-F1 0.9274, test macro-F1 0.9209 on 64 images, 5 errors).
- Highest test macro-F1 (diagnostic): Dropout p=0.3, Dropout p=0.5, Best regularized + scheduled (0.9381).
- Validation macro-F1 spans 0.7825 to 0.9274 across 13 models (spread 0.1449).
- Production test performance: accuracy 0.9219, macro-F1 0.9209, 5 errors.
- Selection protocol: validation macro-F1 only -> resnet18_fine_tuning; 1 model(s) evaluated on the test split.

1. Results overview (validation and test)

Sorted by validation macro-F1. Test columns come from the post-selection diagnostic over the 64 test split.

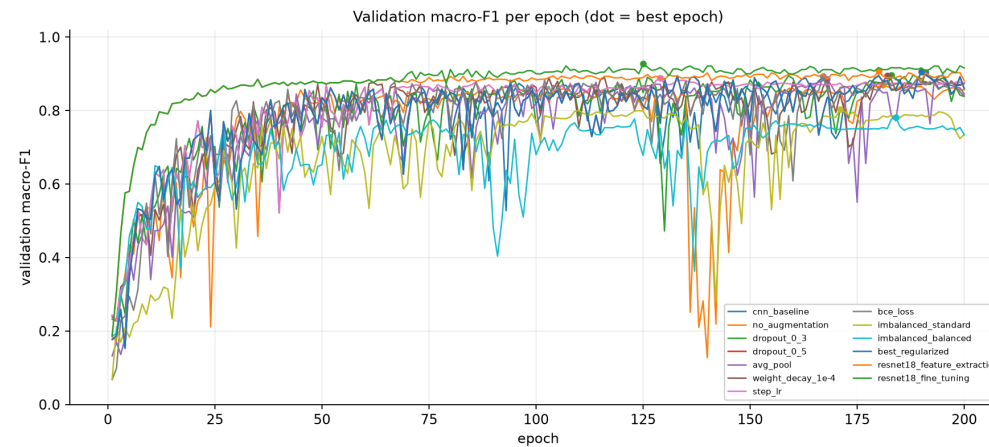
model	category	arch	epoch	val acc	val macro-F1	test acc	test macro-F1	test errors	test mean conf.
ResNet18 fine-tuning	transfer	resnet18	125	0.9267	0.9274	0.9219	0.9209	5/64	0.8892
ResNet18 feature extraction	transfer	resnet18	180	0.9095	0.9072	0.8906	0.8878	7/64	0.8602
Dropout p=0.5	ablation	cnn	190	0.9052	0.9059	0.9375	0.9381	4/64	0.9569
Best regularized + scheduled	main	cnn	190	0.9052	0.9059	0.9375	0.9381	4/64	0.9569
CNN baseline	ablation	cnn	191	0.9052	0.9047	0.9219	0.9213	5/64	0.9430
Dropout p=0.3	ablation	cnn	183	0.8966	0.8966	0.9375	0.9381	4/64	0.9529
Weight decay 1e-4	ablation	cnn	182	0.8966	0.8956	0.8594	0.8652	9/64	0.9446
BCEWithLogitsLoss	ablation	cnn	167	0.8922	0.8930	0.8906	0.8863	7/64	0.8797
StepLR	ablation	cnn	129	0.8879	0.8888	0.8906	0.8931	7/64	0.9422
Average pooling	ablation	cnn	168	0.8836	0.8859	0.8750	0.8752	8/64	0.9379
No augmentation	ablation	cnn	181	0.8707	0.8702	0.8438	0.8435	10/64	0.9098
Imbalanced + standard batches	imbalance	cnn	171	0.8017	0.8032	0.8281	0.8262	11/64	0.8779
Imbalanced + balanced batches	imbalance	cnn	184	0.7845	0.7825	0.7969	0.7962	13/64	0.8889



2. Validation evaluation (232 images, all models)

Metrics at each model's best epoch, selected on the validation macro-F1. ECE is the 10-bin expected calibration error of the raw (uncalibrated) validation probabilities.

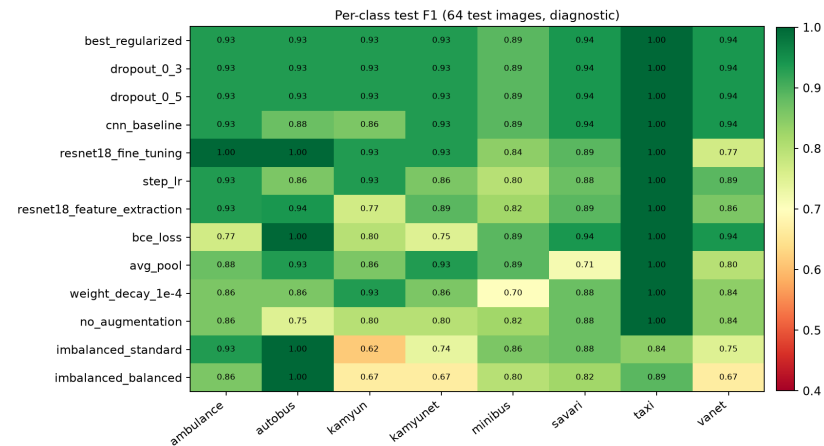
model	loss	epoch	acc	macro-P	macro-R	macro-F1	weighted-F1	ECE	mean conf.	elapsed s
ResNet18 fine-tuning	cross_entropy	125	0.9267	0.9294	0.9287	0.9274	0.9265	0.0636	0.9627	2279
ResNet18 feature extraction	cross_entropy	180	0.9095	0.9112	0.9081	0.9072	0.9082	0.0788	0.8726	2216
Dropout p=0.5	cross_entropy	190	0.9052	0.9137	0.9033	0.9059	0.9056	0.0583	0.9551	2534
Best regularized + scheduled	cross_entropy	190	0.9052	0.9137	0.9033	0.9059	0.9056	0.0583	0.9551	2664
CNN baseline	cross_entropy	191	0.9052	0.9094	0.9043	0.9047	0.9046	0.0813	0.9471	2524
Dropout p=0.3	cross_entropy	183	0.8966	0.9076	0.8950	0.8966	0.8966	0.0845	0.9540	2587
Weight decay 1e-4	cross_entropy	182	0.8966	0.9018	0.8946	0.8956	0.8957	0.0535	0.9494	2737
BCEWithLogitsLoss	bce	167	0.8922	0.8947	0.8922	0.8930	0.8923	0.1122	0.8840	3131
StepLR	cross_entropy	129	0.8879	0.8960	0.8848	0.8888	0.8888	0.0740	0.9431	2678
Average pooling	cross_entropy	168	0.8836	0.8959	0.8819	0.8859	0.8839	0.0812	0.9459	2570
No augmentation	cross_entropy	181	0.8707	0.8726	0.8705	0.8702	0.8703	0.0829	0.9242	2546
Imbalanced + standard batches	cross_entropy	171	0.8017	0.8373	0.8046	0.8032	0.8010	0.1215	0.9050	2706
Imbalanced + balanced batches	cross_entropy	184	0.7845	0.8224	0.7848	0.7825	0.7839	0.1559	0.9218	2397



3. Test evaluation (64 images, diagnostic)

The test split was never used for model selection or tuning; these numbers are a post-selection diagnostic. Ranked by test macro-F1.

model	acc	macro-P	macro-R	macro-F1	errors	mean conf.	below review thr.	top confusion pair	official
Dropout p=0.3	0.9375	0.9472	0.9375	0.9381	4/64	0.9529	7	ambulance/vanet (0.125)	no
Dropout p=0.5	0.9375	0.9472	0.9375	0.9381	4/64	0.9569	6	ambulance/vanet (0.125)	no
Best regularized + scheduled	0.9375	0.9472	0.9375	0.9381	4/64	0.9569	6	ambulance/vanet (0.125)	no
CNN baseline	0.9219	0.9316	0.9219	0.9213	5/64	0.9430	6	ambulance/vanet (0.125)	no
ResNet18 fine-tuning	0.9219	0.9409	0.9219	0.9209	5/64	0.8892	17	savari/vanet (0.250)	yes
StepLR	0.8906	0.9177	0.8906	0.8931	7/64	0.9422	8	autobus/minibus (0.250)	no
ResNet18 feature extraction	0.8906	0.9083	0.8906	0.8878	7/64	0.8602	18	kamyun/kamyunet (0.250)	no
BCEWithLogitsLoss	0.8906	0.8981	0.8906	0.8863	7/64	0.8797	12	ambulance/kamyun (0.125)	no
Average pooling	0.8750	0.8969	0.8750	0.8752	8/64	0.9379	10	savari/vanet (0.375)	no
Weight decay 1e-4	0.8594	0.8982	0.8594	0.8652	9/64	0.9446	6	autobus/minibus (0.250)	no
No augmentation	0.8438	0.8555	0.8438	0.8435	10/64	0.9098	13	kamyun/minibus (0.250)	no
Imbalanced + standard batches	0.8281	0.8486	0.8281	0.8262	11/64	0.8779	18	kamyun/kamyunet (0.250)	no
Imbalanced + balanced batches	0.7969	0.8360	0.7969	0.7962	13/64	0.8889	16	kamyun/kamyunet (0.375)	no



4. Production model in detail: ResNet18 fine-tuning

class	val support	val P	val R	val F1	test support	test P	test R	test F1
ambulance	25	1.0000	1.0000	1.0000	8	1.0000	1.0000	1.0000
autobus	28	0.9655	1.0000	0.9825	8	1.0000	1.0000	1.0000
kamyun	27	0.9130	0.7778	0.8400	8	1.0000	0.8750	0.9333
kamyunet	29	0.7941	0.9310	0.8571	8	1.0000	0.8750	0.9333
minibus	28	0.9655	1.0000	0.9825	8	0.7273	1.0000	0.8421
savari	30	0.9000	0.9000	0.9000	8	0.8000	1.0000	0.8889
taxi	27	0.9259	0.9259	0.9259	8	1.0000	1.0000	1.0000
vanet	38	0.9714	0.8947	0.9315	8	1.0000	0.6250	0.7692

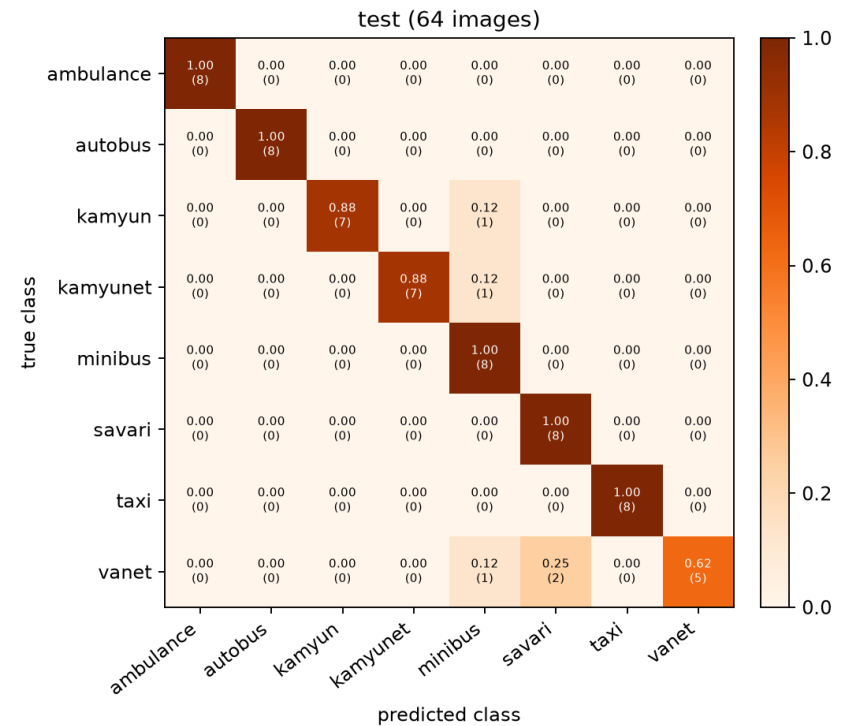
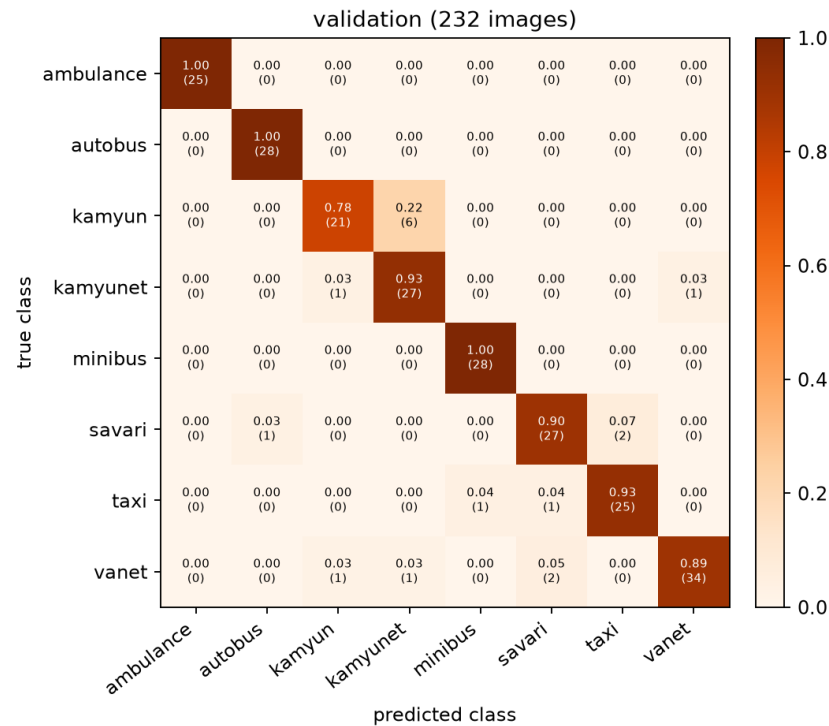
Calibration and uncertainty

split / state	NLL	ECE (10 bins)	Brier	mean confidence	accuracy
validation, raw	0.4426	0.0636	0.1398	0.9627	0.9267
temperature = 2.115	-	-	-	-	-
validation, temperature-scaled	0.3105	0.0438	0.1410	0.9004	0.9267
test, temperature-scaled	0.3109	0.0700	0.1557	0.8892	0.9219

Selective prediction on the test split

field	value
review threshold (selected on validation)	0.8393
accepted / review	47 / 17
automatic coverage	0.7344
review rate	0.2656
automatic accuracy	0.9787
review accuracy	0.7647
errors flagged for review	4 / 5

5. Confusion matrices - production model



Most frequent mutual confusions (test split)

class pair	A as B	B as A	pair confusion
savari <=> vanet	0.0	0.25	0.2500
kamyun <=> minibus	0.125	0.0	0.1250
kamyunet <=> minibus	0.125	0.0	0.1250
minibus <=> vanet	0.0	0.125	0.1250
ambulance <=> autobus	0.0	0.0	0.0000

6. Methodology and caveats

- Model selection used the validation split only (stratified 20% of the training images, seed 42); the best epoch of each run is chosen by validation macro-F1.
- The test split is evaluated after selection, once per model, and is reported here as a diagnostic. It must not be used to pick a model or tune hyperparameters.
- All models share the same class mapping, stratified split and evaluation transform; differences between rows are attributable to the training configuration named by each experiment.
- ECE uses 10 equal-width confidence bins; production-model NLL/Brier are computed after temperature scaling, with the temperature fitted on the validation split.
- Integrity check: recomputed test macro-F1 matches the locked value (abs difference 0.0).
- Per-image test predictions for non-production models are diagnostic outputs of scripts/evaluate_all_models_on_test.py; their per-class F1 values are recomputed from those predictions.

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